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Title: Write a program in C++ to perform following operations on complex numbers Add,Subtract, Multiply, Divide. Use operator overloading for these operations. The objective of this assignment is to learn the concepts of operator overloading.

```
#include <iostream>
using namespace std;

class Complex {
   float real, imag;

public:
   Complex() {
      real = 0;
      imag = 0;
   }

   Complex(float r, float i) {
      real = r;
      imag = i;
   }

   void read() {
      cout << "Enter real part: ";
      cin >> real;
      cout << "Enter imaginary part: ";
      cin >> imag;
   }

   void display() {
      if(imag >= 0)
         cout << real << " + " << imag << "i\n";
```

```
        else
            cout << real << " - " << -imag << "i\n";
    }

    Complex operator+(Complex c) {
        Complex temp;
        temp.real = real + c.real;
        temp.imag = imag + c.imag;
        return temp;
    }

    Complex operator-(Complex c) {
        Complex temp;
        temp.real = real - c.real;
        temp.imag = imag - c.imag;
        return temp;
    }

    Complex operator*(Complex c) {
        Complex temp;
        temp.real = (real * c.real) - (imag * c.imag);
        temp.imag = (real * c.imag) + (imag * c.real);
        return temp;
    }

    Complex operator/(Complex c) {
        Complex temp;
        float denominator = (c.real * c.real) + (c.imag * c.imag);
        temp.real = ((real * c.real) + (imag * c.imag)) / denominator;
        temp.imag = ((imag * c.real) - (real * c.imag)) / denominator;
        return temp;
    }
};

int main() {
    Complex c1, c2, res;
```

```
    cout << "Enter First Complex Number:\n";
    c1.read();

    cout << "\nEnter Second Complex Number:\n";
    c2.read();

    cout << "\n-----------------------------\n";
    cout << "First Number: ";
    c1.display();
    cout << "Second Number: ";
    c2.display();
    cout << "-----------------------------\n";

    res = c1 + c2;
    cout << "Addition: ";
    res.display();

    res = c1 - c2;
    cout << "Subtraction: ";
    res.display();

    res = c1 * c2;
    cout << "Multiplication: ";
    res.display();

    res = c1 / c2;
    cout << "Division: ";
    res.display();

    return 0;
}
```